Biology

Chapter 4:
Biological Molecules

Welcome to AnithUncommon's **Sample Resources!**

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In Biology O'Level, chapter 4

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Yassmin Younes to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology O'Level join our nonprofit!

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This sample is made thanks to, *Yassmin Younes*

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be able to…

- List the chemical elements that make up carbohydrates, lipids, and proteins.
- Describe the structure of large molecules (starch, glycogen, cellulose, lipids, proteins, and DNA) built from smaller basic unit monomers.
- Perform and interpret chemical tests to identify glucose, starch, lipids, proteins, and Vitamin C.
- Understand the biological role of water as a universal solvent in metabolic reactions and digestion.

Reflective Questions to Consider:

Life is an intricately woven web of chemical reactions, all dependent on a handful of carbon-based organic compounds. From the structural rigidity of a giant oak tree to the quick energy needed for a sprinter's pulse, every biological process relies on the exact arrangement of chemical building blocks.

Below are the following core questions you will face as we explore this chapter:

- Since carbohydrates and lipids are both made of Carbon, Hydrogen, and Oxygen, what structural differences allow lipids to store more than double the energy density of carbohydrates?
- If proteins are constructed from just 20 standard amino acids, how can a cell produce millions of completely distinct enzymes, antibodies, and structural components?
- Chemical indicators change color in the presence of specific biological molecules—why is it critical to heat the Benedict's test, while tests for starch, protein, and lipids work at room temperature?

Syllabus Outline:

Unit 4: Biological Molecules

Tip!

- You may find it tricky to remember the specific monomer-polymer pairs and color changes for biochemical tests. The secret is to link the test colors directly to their starting state: Benedict's starts **Blue** (like cold water) and turns **Brick-red** when heated with reducing sugars; Iodine starts **Orange-brown** and turns **Blue-black** when it binds to starch spiraling chains!

Activities in Class:

• **Lesson Breakdown:**

- Perform practical lab experiments testing unknown food samples using Iodine, Benedict's, Biuret, Ethanol Emulsion, and DCPIP reagents to construct a complete food composition profile

Construct molecular models showing how simple glucose units link together to form long starch and glycogen chains.

4.1 Chemical Elements & Large Molecules

The Fundamentals

a. Chemical Composition

→ All living organisms are made of organic compounds containing **Carbon (C)**, **Hydrogen (H)**, and **Oxygen (O)**. Proteins and nucleic acids (DNA) also contain additional elements like **Nitrogen (N)**, **Phosphorus (P)**, and **Sulfur (S)**.

b. Polymers and Monomers

→ Large biological molecules (polymers) are built by joining smaller, repeating units (monomers) together.

Molecule Class	Chemical Elements	Basic Building Blocks (Monomers)	Large Molecule Form (Polymer)
Carbohydrates	C, H, O	Simple sugars	Starch, Glycogen, Cellulose (Polysaccharides)
Lipids (Fats & Oils)	C, H, O	1 Glycerol + 3 Fatty Acids	Triglycerides (Fats / Oils)
Proteins	C, H, O, N (some S)	Amino Acids	Polypeptides / Proteins (e.g., Enzymes, Antibodies)

Mentor's Notes:

Try to **RECOGNIZE** the storage polysaccharides in different organisms.

Plant cells store extra glucose as **Starch**, whereas animal cells store extra glucose as **Glycogen** (mainly in the liver and muscle cells). Both are insoluble polymers of glucose!

In plant cell walls, glucose forms **Cellulose**, which provides structural rigidity.

DNA	C, H, O, N, P	Nucleotides (A, T, C, G bases)	Double-stranded Nucleic Acid

The Structure of DNA

a. The Double Helix

→ DNA (Deoxyribonucleic Acid) is the genetic material present in all cellular organisms.

→ It consists of two long strands coiled around each other to form a **double helix**.

→ Each strand is made of a chain of **nucleotides**. Each nucleotide contains a sugar, a phosphate group, and a nitrogenous base.

b. Base Pairing Rules

→ The two strands are held together by cross-links between complementary base pairs:

• **Adenine (A)** pairs with **Thymine (T)**

• **Cytosine (C)** pairs with **Guanine (G)**

In exams, if asked why cells convert glucose into starch or glycogen for storage, always answer: ***Starch and glycogen are insoluble, so they do not affect the osmotic potential (water potential) of the cell!***

4.2 Food Tests (Biochemical Indicators)

Testing for Nutrients

Target Molecule	Reagent / Test	Initial Color	Positive Result	Key Practical Steps / Details
Reducing Sugars (e.g., Glucose)	Benedict's Solution	Blue	Green → Yellow → **Brick-red** ppt	Add Benedict's reagent and **heat in a boiling water bath (80–100°C)** for 3–5 minutes.
Starch	Iodine Solution	Orange-brown	**Blue-black**	Add a few drops of Iodine solution directly to sample at room temperature.
Proteins	Biuret Reagent	Blue	**Purple / Violet**	Add Biuret solution (Sodium hydroxide + Copper sulfate) and shake gently.
Fats & Lipids	Ethanol Emulsion Test	Clear / Colorless	**Cloudy / Milky-white emulsion**	Dissolve sample in ethanol, pour liquid into distilled water, and shake.

Always pay attention to the exact procedural steps for food tests in **Paper 2 and Paper 6!**

The **Benedict's Test** is the ONLY food test that requires **heating**! If you forget to specify a hot water bath in an exam answer, you will lose the mark.

For the **Ethanol Emulsion Test**, ethanol must be added FIRST to dissolve the fat, and then the liquid is poured into WATER. The white emulsion forms because lipids are insoluble in water and form tiny suspended droplets.

Vitamin C	DCPIP Solution	Dark Blue	**Deformalized / Colorless**	Add sample dropwise to DCPIP solution until the blue color disappears.

4.3 The Role of Water

Water as a Universal Solvent

a. Metabolic Reactions

→ Water is essential for all living organisms because it acts as a **solvent** in which metabolic reactions occur.

→ Enzymes and substrates must be dissolved in water so they can move randomly and collide to react (e.g., in the cytoplasm).

b. Transport Medium

→ Water acts as a transport medium in organisms:

• In animals: Blood plasma is mostly water, transporting dissolved glucose, amino acids, urea, and carbon dioxide around the body.

• In plants: Water carries dissolved mineral ions up through the xylem and transports sucrose and amino acids through the phloem.

c. Excretion

→ Toxic nitrogenous waste products like urea are dissolved in water to be safely excreted as urine.